Hardware Design & Circuit Documentation

mmWave Radar-Based Railway Track Displacement Detection System — RailGuard

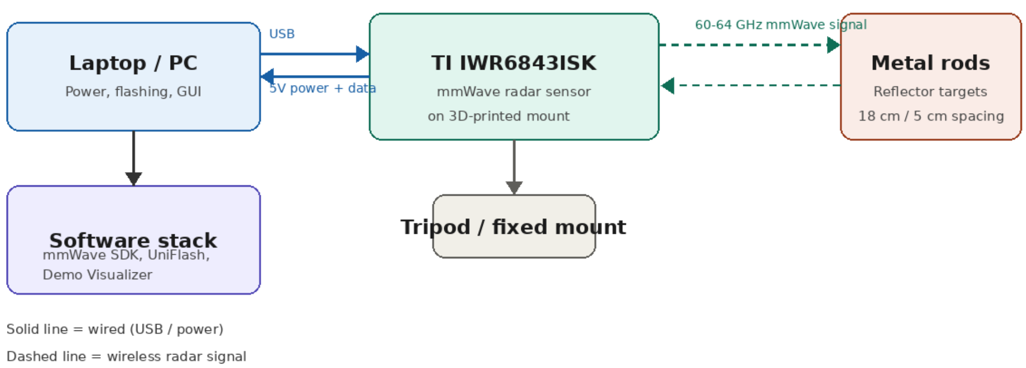
Project overview

RailGuard is a system designed to detect railway track misalignment caused by thermal stress, using a millimeter-wave (mmWave) radar sensor for real-time, non-contact displacement monitoring. Continuously welded rails can buckle or fracture under temperature-driven expansion and contraction, and conventional monitoring methods (strain gauges, manual inspection) are slow, invasive, or reactive. RailGuard addresses this with contactless radar sensing and a live visualization interface.

My role: hardware design

I was responsible for the complete hardware layer of the system: selecting and interfacing the mmWave sensor, ensuring stable, correctly wired power and data connections, and producing full documentation of the interconnection design and every component used — the deliverables shown below.

Hardware interconnection diagram



The TI IWR6843ISK mmWave sensor connects to a host laptop over USB, which supplies both 5V power and the data/flashing interface. The sensor, mounted on a tripod for stable positioning, emits a 60–64 GHz radar signal toward metal rod reflectors standing in for rail displacement points, calibrated at 18 cm from the board with 5 cm spacing between rods. Reflected signals return to the sensor and are processed by the host software stack for real-time visualization.

Component documentation

Component	Role	Key specification

TI IWR6843ISK	mmWave radar sensor, core detection unit	60–64 GHz, 4 GHz bandwidth, integrated FFT/DSP
Tripod / fixed mount	Stable, repeatable radar positioning	Custom-aligned to target distance
Metal rods (reflectors)	Proxy targets simulating track displacement	Calibrated at 18 cm from board, 5 cm spacing
5V power supply / USB	Powers the radar module	Delivered via USB from host
Laptop / PC	Flashing, visualization, debugging	Runs mmWave SDK + Demo Visualizer
3D-printed casing	Protects and secures the sensor	Custom-designed enclosure

Configuration & signal parameters

The sensor configuration file was generated using the TI mmWave Sensing Estimator and validated through the mmWave Demo Visualizer, defining parameters for precise short-range displacement detection: operating frequency 60–64 GHz, 4 GHz bandwidth (~5 cm range resolution), 10 m maximum range, 12 dBm transmit power, and CFAR-based adaptive thresholding for stable static-target detection on rail surfaces.

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